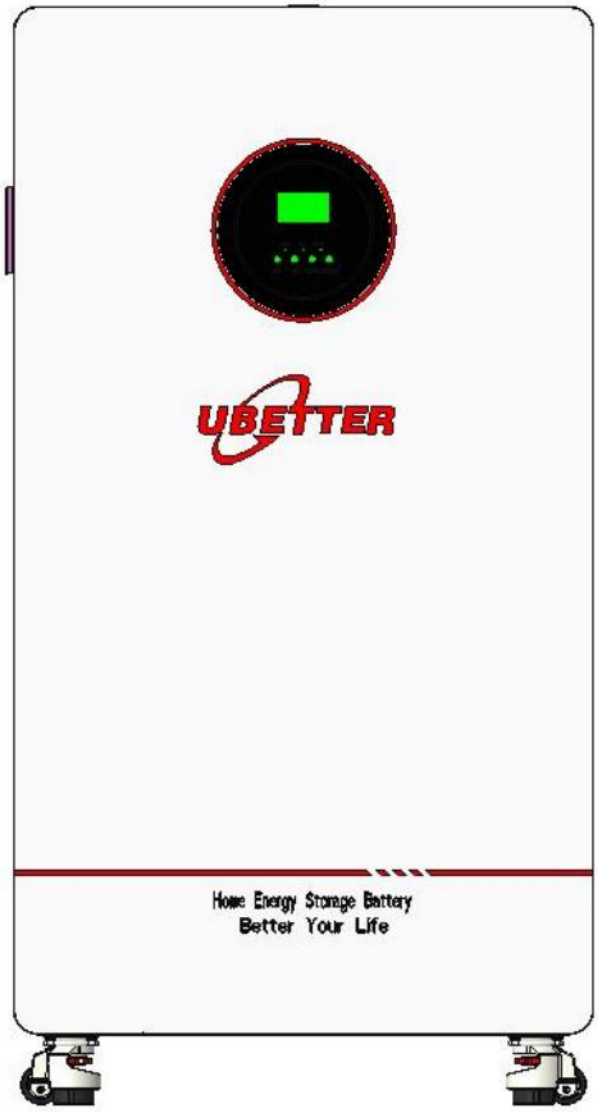


UBETTER TECHNOLOGY Co., Ltd

11KW+15kWh inverter

product specification sheets

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1. scope

This specification applies to LiFePO4 rechargeable battery packs of 48 V 314 Ah15S1P manufactured by UBETTER TECHNOLOGY COMPANY LIMITED.

2. main specifications

2.1 . Battery Cell Specifications TED

	No.	project	general parameters	remark
Cell	1	nominal capacity	314Ah	
	2	nominal voltage	3.2V	average operating voltage
	3	internal impedance	0.17±0.05 mΩ	AC 1 KHz Test (50% SOC)
	4	size	(71.7±0.5) *(173.9±0.8) *(204.3±0.8)	Initial size
	5	weight	5.60±0.15Kg	about
	6	charging cut-off voltage	3.65 V	Cell temperature T > 0℃ Cell temperature T > 0℃
	7	discharge cut-off voltage	2.5 V	Battery temperature T>0℃ Cell temperature T>0℃
			2.0 V	Battery temperature T≤0℃ Cell temperature T≤0℃
	8	charging rate	Standard magnification: 0.5P	25 ℃
	9	discharge rate	Standard magnification: 0.5P	25 ℃
			Maximum continuous discharge rate: 1P	25 ℃
	10	storage temperature	Temperature: -10 ℃~35 ℃, humidity ≤85%	
	11	cycle performance	25℃ 0.5C/0.5C8 000times	Capacity retention ≥70%

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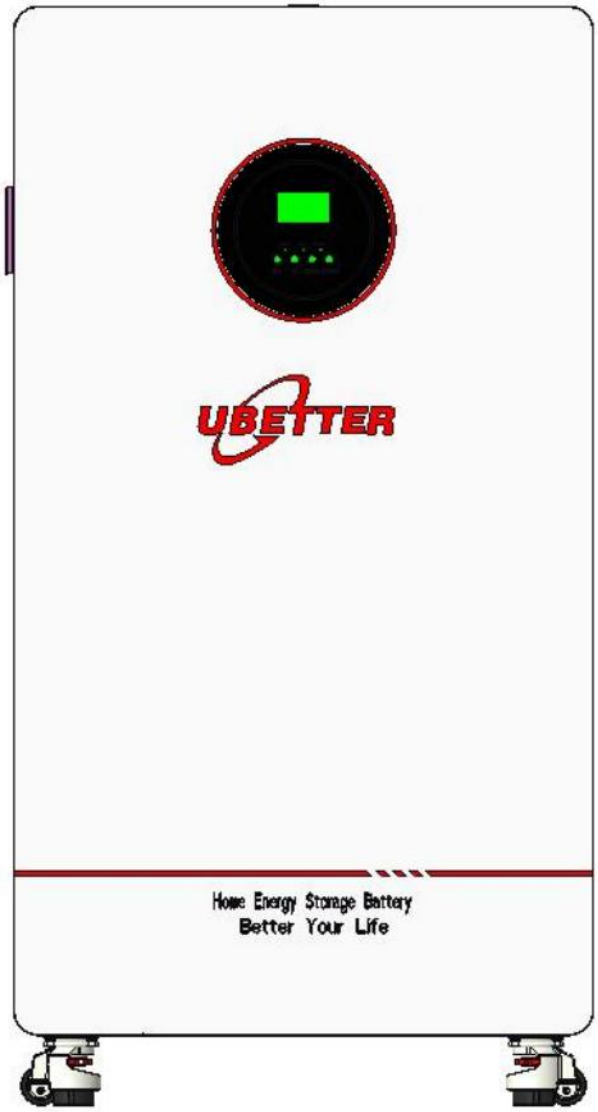
2.2 . Battery pack specifications

	No.	project	general parameters	remark
battery pack	1	combinatorial methods	15S1P	
	2	rated capacity	314Ah	Standard charge after standard discharge (packaging)
	3	working voltage	48V	average operating voltage
	4	Voltage at end of discharge	37.5V	discharge cut-off voltage
	5	charging voltage	54.75V	charging cut-off voltage
	6	internal impedance	≤50mΩ	Internal resistance measured at 1kHz AC, measured after 50% charge must be new batteries with less than 5 cycles within one week of shipment.
	7	standard charging	Constant current 157A constant voltage see no. 5 0.02CA cutoff	Charging time: about 5 hours
	8	Maximum continuous charging current	200A	
	9	standard discharge	Constant current: 157A See No. 4 for termination voltage	
	10	maximum continuous discharge current	200A	
	11	operating temperature range	Charging temperature range: 0~55℃	60±25% relative humidity.
			Discharge: -20 ° C to 65 ° C	
	12	storage temperature range	Less than 12 months: -10 to 35 ℃	60±25% relative humidity, in transit condition.
			Less than 3 months: -10 to 35 ℃	
			Less than 7 days: -20 to 45 ℃	
	13	size	1030*580*234mm	
	14	weight	Approx.: 135 kg	

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3.Appearance and structure size

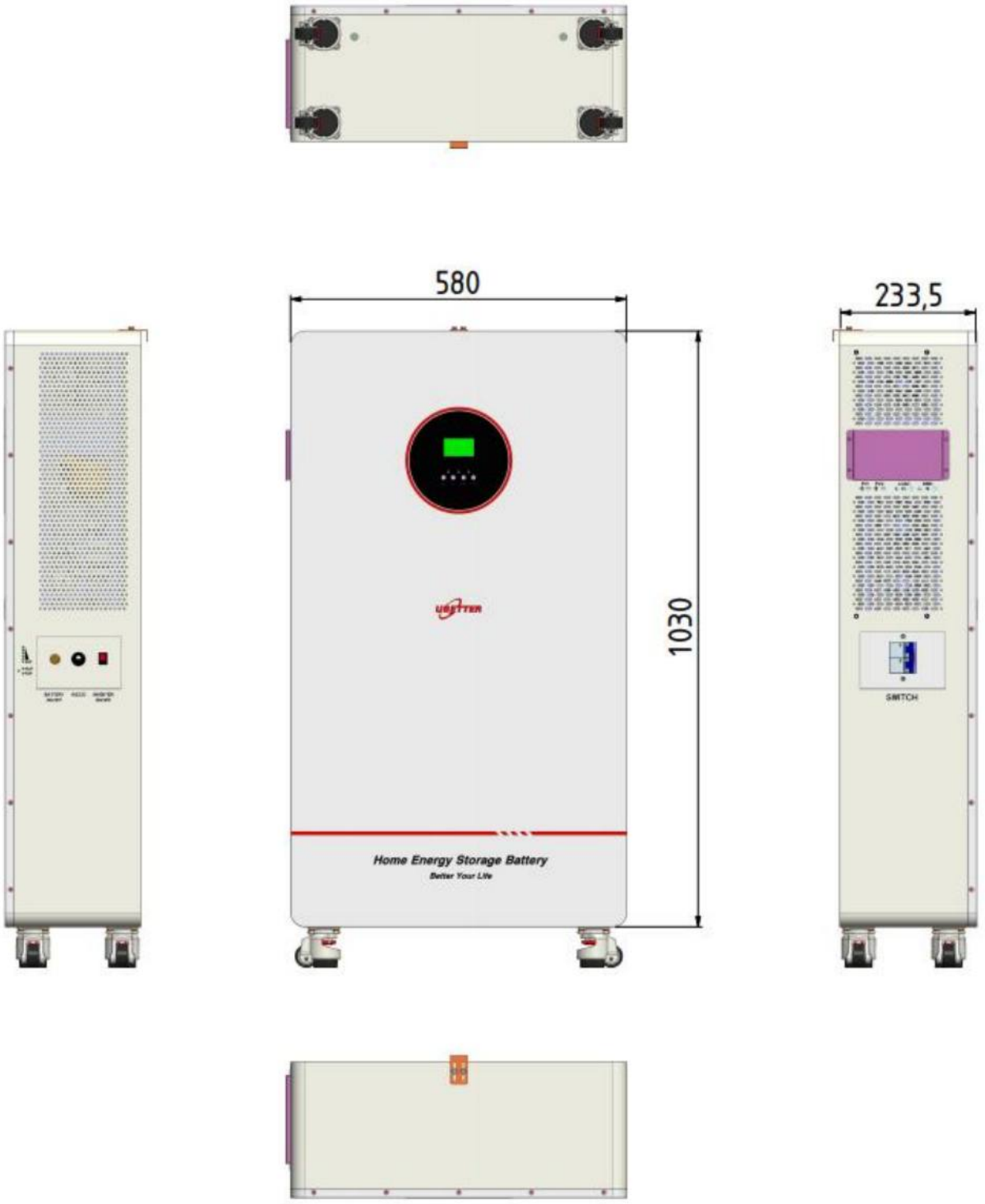
There shall be no scratches, burrs and other mechanical scratches, and the connector shall be free of rust and dirt. For structure and size, please refer to the drawing attached to the battery



L (length)	1030mm	W(width)	580mm	H (height)	234mm
weight	135kg	the largest charge amount	200A	maximum discharge	200A
Unit(mm)					

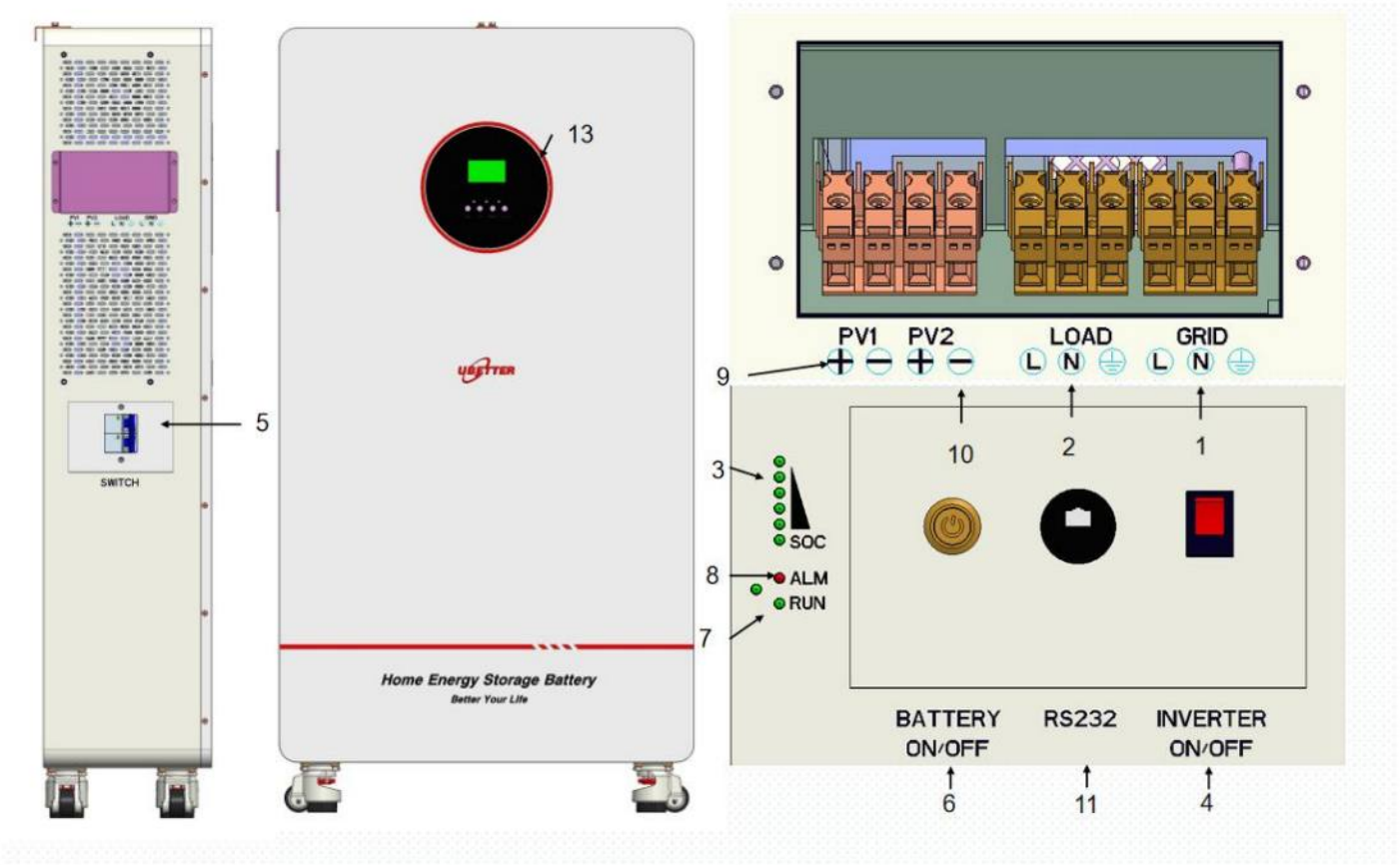
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48V-314Ah15 kWh LiFePO4 Product size diagram



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4.Case Structure Pack of Battery



No.	describe	silk-screen	remark
1	AC input	GRID	output connector
2	AC output	LOAD	output connector
3	power display	SOC	Help users intuitively understand the remaining battery
4	battery switch	BAT/SW	Turn on/off battery power
5	air switch	/	short-circuiter
6	inverter switch	INVERTER/SW	Start/close inverter power supply
7	Running indicator light	RUN	Operation status indicator
8	alert indicator	ALM	fault warning lamp
9	PV input anode	PV+	PV input connector
10	PV input cathode	PV-	PV input connector
11	Inverter wifi stick interface	RS232	Connect inverter wifi stick

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6 Battery test equipment

6.1 size measuring instrument

size measurements shall be made using instruments with an accuracy of not less than 0.1 mm.

6.2 voltmeter

The standard level or more sensitive category specified in the national standard should be used, and the internal resistance should not be less than 10 kiloohms/volt.

6.3 ammeter

Standard levels or more sensitive categories specified in national standards should be used. Total external resistance including ammeter and lead shall be less than 0.01 ohms.

6.4 impedance meter

Impedance shall be measured using the sinusoidal alternating current method (AC 1kHz LCR meter).

7. standard test conditions

The test shall be conducted within one month after the batteries are shipped from our factory and the number of battery cycles before testing shall not exceed five. Unless otherwise defined, tests and measurements shall be carried out at a temperature of 23±2 ° C, a relative humidity of less than 75%, and an atmospheric pressure of 86kPa to 106kPa. Unless otherwise defined, there shall be a 30-minute rest period after charging and a 30-minute rest period after discharging.

8. storage and other

8.1 long-term storage

If stored for a long time (unused, more than three months), batteries should be stored in a dry and cool place. The storage voltage of the battery shall be48 V-49.5V and the battery shall be stored at a temperature of25 ±2℃ and a humidity of 45%-75%. Batteries that have not been used for a long time should be

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recharged every three months to ensure that the battery voltage is within the above range.

8.2 else

Any matters not covered by this specification shall be settled by consultation between Customer and UBETTER.

9. Revisions to this Code

This specification is subject to change upon prior notice from UBETTER.

10. appendix

Lithium-ion rechargeable battery handling precautions and guidelines

introduction

This document, Precautions and Guidelines for Handling Lithium Ion Rechargeable Batteries, applies to battery cells manufactured by UBETTER.

Note (1): If the customer requires other applications or operating conditions not described in this document, the customer should contact UBETTER in advance. Additional experiments may be required to verify performance and safety under these conditions.

Note (2): UBETTER will not be responsible for any accident if the battery is used outside the conditions described in this document.

Note (3): If necessary, UBETTER will inform the customer in writing about improvements to the correct use and handling of batteries.

Danger warning!

- Do not immerse the battery in water or damp it.
- Do not use or store batteries near sources of heat such as fire or heaters.
- Do not use any chargers not recommended by UBETTER.
- Do not reverse the positive (+) and negative (-) terminals.
- Do not connect the battery directly to a wall socket or car cigarette lighter socket.

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- Do not put batteries into fire or heat them directly.

- Do not short-circuit the battery by connecting wires or other metallic objects to the positive (+) and negative (-) terminals.

- Do not puncture the battery case with nails or other sharp objects, and do not crack or trample on it with a hammer.

- Do not knock, throw or apply violent physical impact to the battery.

- Do not solder battery terminals directly.

- Do not disassemble or modify the battery in any way.

- Do not put batteries in microwave ovens or pressurized containers.

- Do not combine batteries with primary batteries (e.g. dry cells) or batteries of different capacities, types or brands.

- Do not use batteries if they smell, generate heat, discolor or deform, or look abnormal in any way. If the battery is in use or charging, remove it from the device or charger immediately and discontinue use.

Warning!

Do not use or store batteries in places exposed to extreme temperatures, such as direct sunlight under car windows in hot weather. Otherwise, the battery may overheat. This also reduces battery performance and/or service life. If the battery leaks and electrolyte gets into your eyes, do not rub it. Instead, rinse with clean, running water and seek immediate medical help. Electrolytes can cause eye injury if left unattended.

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